Detection-Theoretic Decision Rules for AI-Agent Monitors

Overall summary: four ideas over four rounds, a white-box track, decision-only first-stage monitors, three validity audits and a threshold-convention fix

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In one paragraph. AI-control monitoring has improved its *monitors* (LLM judges, prompts, probes) much more than its *decision layer*, the rule that turns per-step scores into audit decisions. Almost every system takes the maximum over steps and compares it with one global honest-run quantile. Starting from detection theory (Neyman–Pearson, CFAR, CUSUM, sequential tests, cascades), we asked five questions:

1. Do accumulating rules beat max at a matched false-alarm budget?
2. Should thresholds be set per context?
3. Can an attacker who knows the detector game it?
4. Can a trusted open model's activations (white-box probes) feed the same decision layer?
5. Can a cheap decision-only model (JEV and similar) triage which steps an LLM monitor reads?

After four rounds, preregistered confirmatory tests, three independent audits and a re-run of every headline under an exact-FPR (conformal) threshold rule, the picture is:

- **max remains the robust default**. Accumulation does not generalize across monitor families.
- **CFAR is the wrong per-context rule** against an attacker who chooses where to act. Damage-equalizing (Stackelberg) allocation beats global at 5%. Its preregistered 1% gain depends on SHADE, and its CI includes 0 under main-task resampling.
- **Detector-aware attacks show no measurable strategy effect**, and execution replay shows that an LLM validity judge does not predict whether an attack works.
- **A white-box probe beats a cheap LLM monitor**: the FIBS follow-up probe read by a trusted Qwen3.5-27B replicated on 1,000 held-out attacks (+0.194 TPR at 1% FPR).
- **Decision-model triage works**:
  - it routes better than nano-LLM and Watcher-style generative triage;
  - a JEV→strong-monitor cascade keeps the strong monitor's detection at half the cost;
  - the same model is much weaker through the decision interface than generatively.

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1 What was done

1. **Literature review.** We read 92 arXiv papers in full and 101 lab and community sources, and wrote separate notes on white-box monitoring and on decision-only models. We ranked 10 ideas, ran the top three, then added a white-box track and Idea 05.
2. **Infrastructure** (private repository `gregkocher/agent-monitor-detection`):
   - loaders for MonitoringBench, Honest-550, the LinuxArena 12-monitor sweep, aksh-n per-step settings, ScaleAI MRT, persistent-state VibeCoding and MOLE;
   - a per-step OpenRouter scoring harness, including the Decisions API;
   - a detector library calibrated at matched false-alarm budgets;
   - a spend ledger;
   - a trusted-reader white-box pipeline and a self-hosted decision model on RunPod GPUs;
   - BashArena execution replay with the official side-task scorers.
3. **Four rounds per idea.** Round 4 was preregistered for Ideas 1–3 and the white-box track. Idea 05 had three preregistered arms and one addendum (Section 4).
4. **Three independent validity audits.** Each re-derived the headline numbers with its own minimal code. They checked calibration leakage, grouping, confounds, onset labels, attack validity, spend matching, selection effects, preregistration timing, truncation and reasoning leakage. Corrections are tracked claim by claim in `results/fixes/`, `results/audit2/` and `results/audit3/corrections.csv`.
5. **Threshold-convention fix.** The third audit found that the plug-in randomized threshold used everywhere runs one calibration point hot for continuous scores (Section 5). Every headline was re-run under a conformal randomized rule with exact expected FPR (`results/fixes2/headline_threshold_rerun.csv`, 57 rows). Seven conclusions changed; the reports and the figure below use the corrected numbers.

2 Most important and most novel findings

1. **A white-box probe on a trusted reader beats a cheap LLM monitor, confirmed** [C]. The FIBS axial probe, read on a per-step follow-up question by a trusted Qwen3.5-27B, raises TPR at 1% trajectory FPR over `gpt-4.1-mini` by +0.194 [+0.089, +0.288] on 1,000 held-out MonitoringBench attacks (0.26 vs 0.06). With both monitors bisected to exactly 1.00% FPR the gain is +0.164. The pilot found the same (+0.204), and probe triage plus the LLM gains +0.123 while calling the LLM on 13% of steps [E]. The comparator is a cheap monitor, and the replication covers attack structures S1–S3 only.
2. **Decision-model triage** (Idea 05).
   - *It beats cheap generative triage at equal escalation.* JEV beats a `gpt-4.1-nano` first stage by +0.170 [+0.131, +0.201] on aksh-n [C]. This compares first-stage designs: JEV saw a three-step window, nano did not. On MonitoringBench into generative GPT-6 Luna, JEV beats a Watcher-style GPT-5.6 Luna severity stage by +0.093 [+0.041, +0.130] ($e = 10\%$) and +0.049 [+0.016, +0.104] ($e = 20\%$) [C]. That stage used Watcher's published rubric, not Watcher Live. The Watcher-style triage call also costs more than running generative Luna on every step.
   - *A cascade into a strong monitor keeps its detection on MonitoringBench.* JEV→generative Luna at $e = 20\%$ is non-inferior to Luna on every step (0.432 vs 0.381, +0.051 [−0.004, +0.106]) [C], at about half the cost ($6.6 vs $13.0 per 1,000 trajectories). Superiority is not shown.
   - *With the dataset's Gemini monitor, the pooled non-inferiority is a cancellation.* The pooled estimate is +0.005, made of SHADE +0.18 and BashArena-opus-4-6 −0.22. It is not non-inferior without SHADE or at 5% FPR.
   - *The interface penalty.* The same model, GPT-6 Luna, is far weaker through the decision interface than generatively: standalone TPR 0.17 vs 0.33 pooled over three panels (−0.158 [−0.227, −0.113]) and 0.04 vs 0.38 on MonitoringBench [C]. Luna Decisions also routes worse than JEV (−0.034) [C].
3. **Stackelberg vs CFAR, and the SHADE dependence** (Idea 2). CFAR, which equalizes FPR per context, is the worst allocation against an attacker who picks the context: −0.280 at 1% with exact spend matching [E]. Stackelberg (damage-equalizing) beats global at 5% (+0.076 [+0.026, +0.124]) [E]. The preregistered 1% test was met, +0.107 [+0.024, +0.152] [C], but the informed attacker picks SHADE (5 main tasks) in every split. With main-task resampling the CI is $[-0.003, +0.251]$.
4. **Max is hard to beat** (Idea 1 and the white-box track). Textbook sequential tests lose to max-type rules on almost every set. The preregistered cross-session test pooled over nine monitor families gives top-3 sum − max −0.066 [−0.147, +0.015] [C]. No rule beats max on any of 8 MOLE corpora [C], no rule chooser beats always-max out of sample, and max is also best on every probe stream. Positive cases exist (GPT-4o +0.134 [+0.024, +0.217]) but are monitor- and read-out-specific.
5. **Evaluation pitfalls that changed conclusions** (Section 5): the plug-in threshold running hot at low FPR; label leakage through reasoning and length; truncation asymmetry between monitors; pooled non-inferiority hiding per-setting failures; trajectory vs task resampling; and LLM-judged attack validity vs execution.

3 Results at a glance

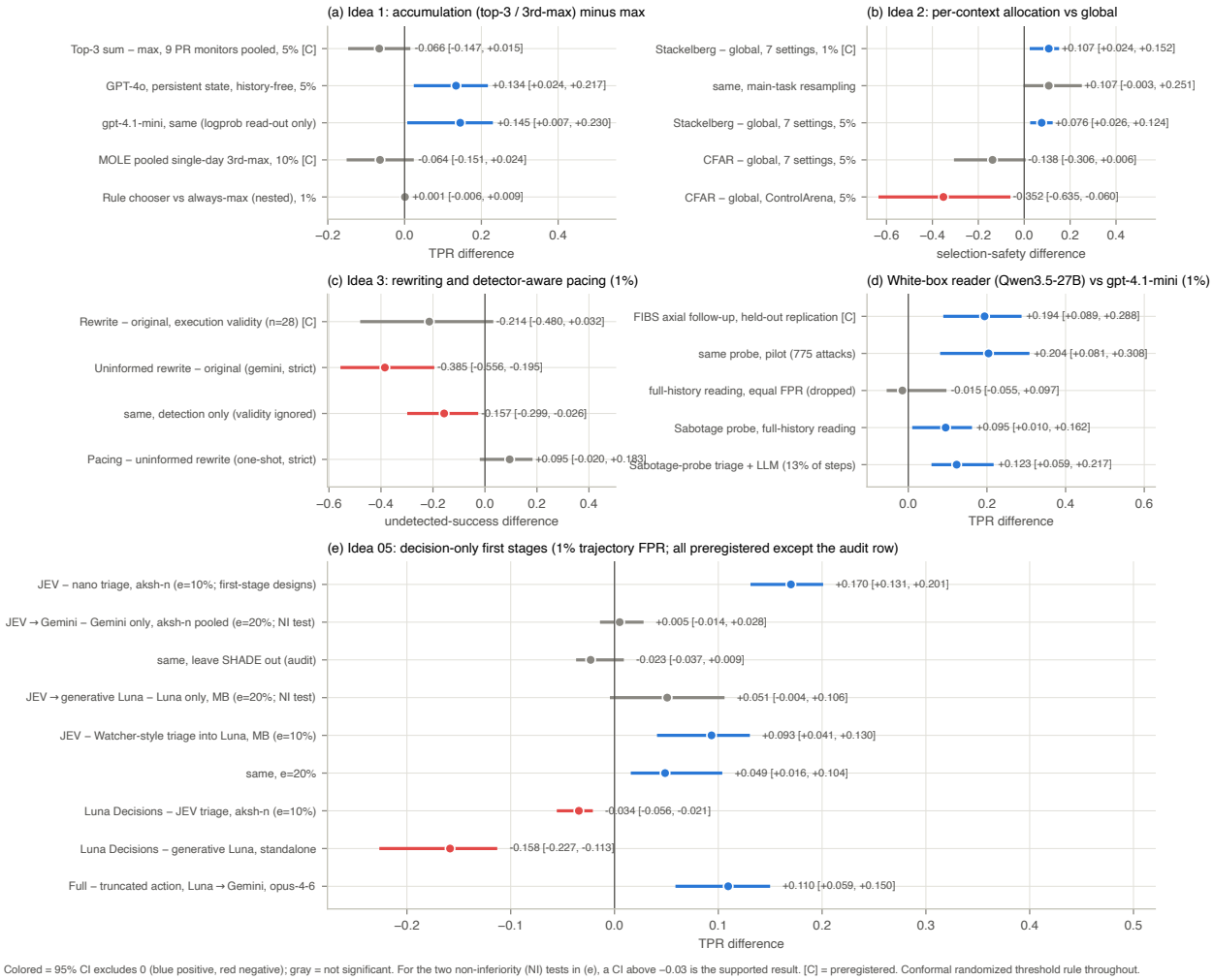


Figure 1: Headline effects with corrected 95% CIs under the conformal threshold rule. Colored = CI excludes zero; gray = not significant. Panels (a)–(d) and the first three rows of (e) are read from `results/fixes2/headline_threshold_rerun.csv`; the remaining rows of (e) from `results/r5_luna/` and `results/r5_watch/` (`make_overview_fig.py`). Panel (c), third row: detection only, with rewrite validity ignored (it was mislabeled "valid rewrites" in the October 22 version).

4 The five research directions

4.1 Idea 1: accumulation vs max at a matched budget

Ran. Round 1: 40 existing stream sets and a controlled history experiment on MonitoringBench. Round 2: cross-session persistent state with four monitors, MOLE, a rule chooser. Round 3: all 8 MOLE corpora under one monitor, unreset streams, re-flagging monitor designs. Round 4 (preregistered): persistent state re-scored by nine monitor families with identical inputs, MOLE with a stronger day monitor (GPT-6 Luna), unreset streams with CIs.

Holds.

- Max, 2nd-max and top-3 sum are strong defaults; textbook sequential tests lose almost everywhere [E].
- Cross-session accumulation does not generalize: pooled over nine monitors, $-0.066\,[-0.147, +0.015]$ [C]. After Holm only GPT-6 Luna's loss (-0.42) is significant.
- MOLE: no accumulation rule beats max on any of 8 corpora ($-0.064\,[-0.151, +0.024]$) [C]. A stronger day monitor does not create a gain ($+0.047\,[-0.026, +0.125]$) [C].
- No rule chooser beats always-max out of sample ($+0.001\,[-0.006, +0.009]$) [E].

Weakened.

- GPT-4o history-free cross-session gain $+0.134\,[+0.024, +0.217]$ [E]: real, but one monitor.
- `gpt-4.1-mini`'s $+0.145\,[+0.007, +0.230]$ appears only with its logprob-expected read-out. The integer read-out of the same calls gives -0.03 (audit 3).
- The unreset-stream CUSUM result (0.72 vs 0.59) holds as computed, but with CIs it is $+0.132\,[-0.086, +0.296]$ [C]. No forgetting rule beats max pooled over nine monitors.

Dropped. "Accumulators gain more with history" (a prompt-instruction confound); the round-2 MOLE gain; the repeat-spike mechanism (not predictive across monitors, [C]); "Luna's insider signal is account-level" (it comes from attack days).

4.2 Idea 2: per-context thresholds

Ran. Round 1: heterogeneity measurement, budget-matched allocation (global, CFAR, Mondrian, partial pooling, max-detection, Stackelberg), context-selecting and poisoning attackers. Round 2: transfer, multi-context poisoning, severity, probing attackers, white-box panels. Round 3: Watcher-like severity tiers. Round 4 (pre-specified): allocation at $B = 1\%$ with many honest runs per context and exact spend matching.

Holds.

- Heterogeneity is real across deployments, small within one, and large for probes (τ up to 3.79 vs ≤ 0.46 for LLM monitors) [E].
- CFAR is the worst rule against context selection: -0.280 at 1% with exact spend, worst at every budget. A few alarm-feedback probes exploit it [E].
- Stackelberg beats global at 5% ($+0.076\,[+0.026, +0.124]$) [E].
- Per-context rules have a spend floor of about $\sum_c w_c/(n_c + 1)$, so they need roughly $1/B$ calibration runs per context [E].
- A frozen trusted reference resists poisoning but over-spends under genuine drift; Watcher-like fixed severity tiers lose 0.06–0.12 to a plain suspicion threshold [E].

Weakened. The preregistered Stackelberg gain at 1% ($+0.107\,[+0.024, +0.152]$) [C] is driven by SHADE, and with main-task resampling the CI is $[-0.003, +0.251]$. The matching uses held-out honest data as the operating point: fair across rules, but not deployable. Fit on other side tasks, Stackelberg loses to global.

Dropped. "CFAR doubles probe TPR at 1%" (not estimable; 1.3–1.5× at equal spend); the ControlArena 1% gain.

4.3 Idea 3: detector-aware attackers

Ran. Round 1: a `gpt-4.1` rewriter told the monitor prompt and detector. Round 2: a strong monitor, pacing attackers, an uninformed-rewrite control. Round 3: a different-family rewriter, a fabricated-output leak test, realized FPR, replay feasibility. Round 4 (preregistered): originals and rewrites replayed in BashArena with the official side-task scorers.

Holds.

- No detectable effect of detector knowledge: pacing − plain rewrite $+0.095\,[-0.020, +0.183]$ (round 2, strict validity) [E] and $-0.107\,[-0.357, +0.120]$ under execution validity [C].
- Max, CUSUM $k = 1.5$ and top-3 sum hold nominal FPR; the multi-chart OR runs 1.7–1.9× nominal [E].
- Pacing costs validity, not detection: $-0.321\,[-0.625, -0.034]$, of which -0.24 is broken attacks [E].

Weakened. "Any LLM rewrite makes attacks easier to catch." With judge or static validity the drops were large and significant (gemini rewrite $-0.385\,[-0.556, -0.195]$; detection only $-0.157\,[-0.299, -0.026]$). Under execution validity the preregistered contrast is $-0.214\,[-0.480, +0.032]$ ($n = 28$) [C], and $-0.395\,[-0.667, -0.083]$ without firewall attacks [E]. No execution-validity cell survives Holm. The direction is consistent; significance is not established.

Also found. 24 of 33 "successful" originals reproduce in BashArena 1.1, and 8 of the 9 failures are firewall attacks (environment drift). Judge-invalid rewrites succeed 4 of 5 and 6 of 7 times.

Retracted. Round 1's "aware attacks backfire" (no control) and round 2's "+0.21 pacing gain" (adaptive vs one-shot arms).

4.4 White-box track

Ran. A trusted Qwen3.5-27B reader on H200 pods, reading per step or with full history: a synthetic sabotage probe, Apollo's deception probe, FAR.AI FIBS probes on step tokens and on a follow-up question, and a J-lens watch-list, all through the same decision layer. Round 4 (preregistered): replication on 1,000 new MonitoringBench attacks; regenerated LinuxArena streams with paired tests; a MOLE policy-activation arm with account-matched benign sessions.

Holds.

- The FIBS axial follow-up probe beats `gpt-4.1-mini`: $+0.194\,[+0.089, +0.288]$, and $+0.164$ at exactly 1.00% FPR for both [C].
- Sabotage-probe triage plus LLM: $+0.123\,[+0.059, +0.217]$ at matched realized FPR with about 87% fewer LLM calls [E].
- LinuxArena: the full-reading FIBS probe is Holm-significantly above 6–7 of 12 sweep monitors at nominal 1% and below `claude-opus-4-6` [C]. Counts at exactly equal 1% FPR are rule-dependent (one calibration point).
- MOLE with matched negatives: no AUROC difference between policy and reader, but at 5% session FPR the reader detects more (policy − reader -0.24), and the insider instruction lowers the policy probe's TPR (-0.06) [C].

Weakened or dropped. The full-history sabotage probe ($+0.095$; Holm p 0.22, opus-4-5 only). The full-history FIBS axial gain at equal FPR is dropped ($-0.015\,[-0.055, +0.097]$; the earlier $+0.061$ sat at a plug-in FPR floor). J-lens watch-lists are weak monitors (AUROC ≤ 0.72).

4.5 Idea 05: decision-only models as first-stage monitors

Ran. JEV 1.14 (TypeSafe; label probabilities only, about $0.04 per million input tokens) scored every step of MonitoringBench (511 honest, 1,637 attacks) and seven aksh-n settings: 421k held-out steps. The cascade escalates the top-e share of steps to a cached per-step monitor; trajectory max; fixed trajectory FPR. Three preregistered arms:

- round 1 (H1–H3) with nano, white-box probe, length, random and oracle gates;
- GPT-6 Luna Decisions vs JEV, vs nano, vs generative GPT-6 Luna, and full vs truncated actions (H5a–H5d), plus an addendum with generative Luna as MonitoringBench's strong second stage;
- a Watcher-style GPT-5.6 Luna severity stage (W1–W3).

Also: an exploratory length, format and truncation check, and self-hosted Perplexity Decider v1.2.

Holds (1% trajectory FPR).

- JEV triage beats nano triage on aksh-n by +0.170 [+0.131, +0.201], and Luna Decisions beats it by +0.130 [C]. Both compare first-stage designs (window and rubric vs a history-free prompt).
- JEV beats Watcher-style triage into generative Luna on MonitoringBench by +0.093 / +0.049 ($e = 10/20\%$) [C]. Luna Decisions and Decider vs Watcher-style are not significant after Holm.
- JEV→generative Luna is non-inferior to Luna on every step (+0.051 [−0.004, +0.106]) [C].
- The interface penalty: Luna Decisions vs generative Luna, −0.158 pooled and 0.04 vs 0.38 on MonitoringBench [C]. Luna Decisions routes worse than JEV on aksh-n (−0.034 [−0.056, −0.021]) [C].
- Full untruncated actions raise BashArena-opus-4-6 cascade TPR (+0.110 for Luna [C]). For JEV, 0.49→0.76 matches Gemini alone (0.71) [E].
- Confident JEV misses are shared with the LLM (1.1–5.9× the base miss rate), so confidence gating cannot rescue them [E]. The open Decider model routes as well as JEV [E].

Qualified.

- H1-AK (JEV→Gemini non-inferior to Gemini, +0.005 [−0.014, +0.028]) is a cancellation of SHADE +0.18 and opus-4-6 −0.22. It is not non-inferior without SHADE (−0.023) or at 5% FPR (−0.025).
- H1-MB and H3-MB use a degenerate weak second stage (mini TPR 0.10). H2-MB is not supported.
- Standalone JEV comparisons are at unequal FPR: JEV cannot get below about 2.2% realized FPR on MonitoringBench, and on aksh-n Gemini saw full text.

Not done. Attacks that target the decision model; whether such a first stage is robust to an adaptive attacker remains open.

5 Evaluation pitfalls

These are methodological findings in their own right. Each changed at least one conclusion in this project.

1. **The plug-in threshold runs hot at low FPR.**
   - The common randomized-quantile rule realizes $(\lfloor \alpha n \rfloor + 1)/(n + 1)$ for continuous scores:

1.98% at $n = 100$ and $\alpha = 1\%$, 1.49% at $n = 200$. Randomizing at ties only fixes integer-score monitors.

- Continuous probes were therefore compared with tied LLM scores at unequal realized FPR. Under a conformal rule with exact expected FPR, the full-history FIBS gain vanished ($+0.061 \to -0.015$), and LinuxArena win counts moved.
- Report realized FPR and compare monitors at equal realized FPR.

2. **Reasoning and length leak the label in public benchmarks.**
   - Agent reasoning length alone separates MonitoringBench honest from attack runs with AUROC 0.03 (honest runs have reasoning on every step, attacks on 43%), and aksh-n LinuxArena with 0.81. No monitor we scored saw reasoning; externally computed scores cannot be checked.
   - Length alone reaches AUROC 0.84–0.95 on the BashArena sets and 0.90 on agentdojo, and step count 0.70 on MonitoringBench.
3. **Truncation asymmetry.** Our pipeline keeps the head and tail of actions over 4,000 characters, while the datasets' own monitors scored full text. On BashArena-opus-4-6 attacks lose a median 52% of action text, and the external monitor's top step is a truncated one in 75% of them. Comparisons across monitors are only like-for-like when both saw the same text.
4. **Pooled non-inferiority hides per-setting failures.** H1-AK passes pooled only because a SHADE gain cancels an opus-4-6 loss. Report per-setting, leave-one-out and equal-weight versions next to any pooled test.
5. **Trajectory vs task resampling.** Settings with few main tasks (SHADE 5, iac 5, agentdojo 16) look precise when resampled by trajectory. The Idea 2 1% primary goes from $[+0.024, +0.152]$ to $[-0.003, +0.251]$ with main-task resampling. Resample the unit that was sampled.
6. **LLM-judged attack validity is not execution.** In BashArena replay the judge's validity label does not predict success (judge-invalid rewrites succeeded 4 of 5 and 6 of 7 times), and non-reproductions were mostly environment drift. Judge or static validity produced significant "rewrites make attacks easier to catch" effects that execution validity does not establish.
7. **Read-out choice.** The same `gpt-4.1-mini` calls give a +0.14 accumulation gain with a logprob-expected score and −0.03 with the generated integer.

6 Recommendations for practitioners

1. **Default trajectory rule:** max, with a length-aware null where lengths differ; top-3 sum or CUSUM with $k \approx 1.5$ as alternatives. Calibrate any OR combination jointly. Use a threshold rule with exact FPR and verify realized FPR on held-out honest runs.
2. **Accumulate across sessions only after checking** for your monitor, read-out and data; it does not generalize across monitor families.
3. **Per-context thresholds:** measure heterogeneity first; never use CFAR when attackers can choose contexts; use Stackelberg allocation fit on threat-like attack data with shrinkage; budget about $1/B$ honest runs per context; use a frozen trusted reference against poisoning.
4. **Severity is not suspicion.** Watcher-style severity tracks stakes, and on BashArena-style tasks most honest sessions contain high-severity steps. Validate tiers against an honest hit rate before wiring them in.
5. **Cheap triage:** a windowed decision-only model or a white-box probe in front of a strong LLM monitor keeps detection at a fraction of the cost. Gate on suspicion, not confidence; give the

first stage full text or add a length gate; do not use the decision model alone as the monitor.

6. **Red-teaming with LLM rewrites** needs an uninformed-rewrite control and execution-grounded validity.
7. **White-box:** a probe read on a follow-up question is a cheap complement to LLM monitors and an efficient triage stage. Calibrate probe thresholds per context.

7 What is new (to our knowledge, per the literature review)

- A matched-budget comparison of sequential and anytime-valid statistics vs max on real AI-control score streams, including a preregistered test across nine monitor families. It answers MonitoringBench's stated open question, largely in favor of max.
- A per-context threshold study for control monitors: CFAR's failure under attack selection, the Stackelberg alternative, the spend floor, poisoning-robust baselines and Watcher-like tiers.
- Evidence that LLM rewrite artifacts dominate detector-aware red-teaming, and an execution-replay test of attack validity.
- A trusted-reader white-box pipeline through the same decision layer, with a preregistered replication of a probe beating a cheap LLM monitor.
- The first executed cascades with a decision-only first stage on agent-sabotage trajectories at fixed trajectory-level FPR, a same-model interface comparison, a Watcher-style triage comparison, and an open decision model in the same pipeline.

8 Validity process and errata

Audit 1 found 2 critical, 9 major and 13 minor issues, and audit 2 found 4 major and 10 minor. Audit 3 (round 4 and Idea 05 round 1) lists 72 rows, led by the SHADE dependence of two primaries, firewall drift in replay, omitted preregistered analyses, read-out artifacts, FPR-mismatched wins, the threshold convention and truncation. Every headline number reproduced with independent code. Main changes relative to the October 22 summary:

Claim (Oct 22)	Now	Verdict	Source
Cross-session GPT-4o $+0.146\,[0.025, 0.226]$	$+0.134\,[+0.024, +0.217]$ (conformal); 9-monitor pooled -0.066	holds, does not generalize	fixes2, r4_cs
gpt-4.1-mini cross-session $[-0.001, 0.228]$	$[+0.007, +0.230]$, but only with logprob read-out (integer -0.03)	weakened	audit3, fixes2
Unreset CUSUM beats max (0.71 vs 0.58)	$+0.132\,[-0.086, +0.296]$	not significant	r4_cs
Stackelberg > global at 5%	+0.076; 1% primary met but SHADE-driven ($[-0.003, +0.251]$ by task)	holds / fragile	r4_ctx, audit3
Any rewrite makes attacks easier to catch	execution validity $-0.214\,[-0.480, +0.032]$	weakened	r4_replay
Overview row "valid rewrites" -0.143	detection only, validity ignored: $-0.157\,[-0.299, -0.026]$	relabeled	fixes2
FIBS follow-up probe beats mini (pilot)	replicated on held-out attacks, +0.194	confirmed	r4_wb
Full-history FIBS probe lead	-0.015 at equal FPR (plug-in floor)	dropped	fixes2
LinuxArena reader scores lost	regenerated; 6–7 of 12 above, below opus-4-6	replaced	r4_wb
MOLE: policy does not beat reader	reader detects more at 5% FPR (-0.24)	corrected	audit3

9 Budget, compute and artifacts

Item	Spend	Notes
Idea 1 (OpenRouter)	$562.34	rounds 1–4: $109.56, $147.89, $161.36, $143.53
Idea 2	$38.93	rounds 1 and 3 ($17.11, $21.82); rounds 2 and 4 used cached scores
Idea 3	$497.41	rounds 1–4: $68.36, $304.77, $111.02, $13.26; replays ran locally
White-box track	$37.42	pilot $2.95, round 4 $34.47
Idea 05	$303.70	dev $5.34, round 1 $82.08, Luna arm $154.38, Watcher arm $61.90
Audit 1 re-scoring	$25.31	audits 2 and 3 and the threshold re-run: $0
Setup	$0.53	
OpenRouter total	**$1,465.64**	per-call ledger outside the repository; it does not record which account paid
RunPod GPU	≈$207	white-box pilot ≈$70, round 4 ≈$70, Decider ≈$67

All pods were on-demand on the MATS RunPod account (pilot: two H200; round 4: three H200 and one H100; Decider: one H100) and were deleted after outputs were copied off and verified.

- **Repository:** https://github.com/gregkocher/agent-monitor-detection (private). Code in `amd/` and `tracks/`, results in `results/`, write-ups in `docs/`; full dated history in `EXPERIMENT_LOG.md`.
- **Detailed reports** in `docs/reports/`:
  - `01_Idea1_Matched_Budget.pdf`
  - `02_Idea2_Per_Context_Thresholds.pdf`
  - `03_Idea3_Aware_Attacker.pdf`
  - `04_WhiteBox_Pilot.pdf`
  - `05_Idea5_Decision_Model_Monitors.pdf`
- **Literature:** `docs/research/` has the literature summary, idea ranking, reading cards and research notes.

10 Limitations and next steps

Limitations.

- **Offline replay of public benchmarks.** Almost everything runs on fixed, already-generated trajectories; only Idea 3 round 4 executed attacks. The benchmarks may be in the training data of the models used as monitors.
- **Weak comparators and small samples.** Several confirmed effects are against cheap monitors (`gpt-4.1-mini`). Honest sets of 200–550 runs put 1% FPR on a handful of calibration exceedances, and several settings have few main tasks.
- **External scores.** The aksh-n Gemini and LinuxArena sweep scores are the dataset authors' outputs, computed on full text with inputs we cannot verify.
- **Closed, recent model versions** (JEV 1.14, GPT-6 Luna Decisions in beta, GPT-5.6 Luna) may not transfer to later versions.
- **No adaptive attacker against probes or decision-model first stages.** Attacks targeting the decision model were not run; this remains open.

Next steps, in priority order:

1. **Strong comparators everywhere:** a frontier LLM monitor on the white-box MonitoringBench comparison, and strong second stages on all Idea 05 panels.
2. **Full-text first stages:** re-run H1-AK with full-text JEV and a length gate, and separate the interface effect from prompt differences in H5c.
3. **Adaptive attackers** against probes and decision-model first stages, with an uninformed-rewrite control and execution validity.
4. **Stackelberg at 1% with more main tasks**, so that the result does not rest on one 5-task setting.
5. **Fresh on-policy rollouts** with reasoning retained, so policy and reader arms see identical text.